GR&R Study Report

Gauge Repeatability & Reproducibility — AIAG MSA 4th Edition

Equipment / Gage:	Mitutoyo 293-340-30 Digital Micrometer SN-00142
Study Performed By:	J. Martinez, Senior Quality Engineer
Report Date:	2026-05-11
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 | AIAG MSA 4th Ed.

Variation Components Summary

Component	Study Variation	% of TV	Threshold
EV — Equipment (Repeatability)	0.0087	10.1%	—
AV — Appraiser (Reproducibility)	0.0119	13.7%	—
GR&R — Combined	**0.0148**	**17.0%**	**≤ 10%: Accept | ≤ 30%: Marginal**
PV — Part Variation	0.0854	98.5%	—
TV — Total Variation	0.0867	100.0%	—
ndc — Number of Distinct Categories	8.2	—	≥ 5 required

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	9.9945	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.0029	Mean of operator R-bars
Operator Mean Spread	X-diff	0.0044	max(X-bar op) - min(X-bar op)
Part Range	Rp	0.0527	max(part avg) - min(part avg)

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	9.9938	0.0033
Bob	9.9972	0.0023
Carol	9.9927	0.0030

Acceptance Determination

[CAUTION] MARGINAL

%GR&R = 17.0% | ndc = 8.2

The measurement system is MARGINAL. %GR&R; falls between 10% and 30%. Use with caution — approval should be based on application importance, cost of gage improvement, and customer concurrence. Investigate dominant variation source (EV vs. AV) and consider operator re-training or gage upgrade.

ndc = 8.2 — ADEQUATE: The gage can resolve 5 or more distinct categories of part variation.

AIAG Constants Applied — K1=? (10 trials), K2=2.7 (3 operators), K3 per part count. Study variation = 5.15σ (99% of normal distribution).

Reference: AIAG Measurement Systems Analysis Reference Manual, 4th Edition (2010). Regulatory basis: 21 CFR Part 820.72 — Inspection, Measuring, and Test Equipment (FDA Quality System Regulation). This report is a controlled quality record. Retain per applicable document control procedures.